YEAR 7 MATHS · CHAPTER 2 · MIND MAP

Number properties and patterns

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1 Factors and multiples

Factors go into a number. Multiples come out of its times table.

Factors of 12: 1, 2, 3, 4, 6, 12

Multiples of 4: 4, 8, 12, 16...

2 HCF and LCM

Highest common factor: the biggest number into both. Lowest common multiple: the smallest both go into.

12 and 18: HCF is **6**, LCM is **36**

3 Divisibility tests

By 2: even. By 3: digits add to a multiple of 3. By 5: ends in 0 or 5. By 9: digits add to a multiple of 9.

126: digits add to 9, so it divides by 3 and 9

4 Prime numbers

A prime has exactly two factors: 1 and itself.

2, 3, 5, 7, 11, 13, 17, 19

1 isn't prime. 2 is the only even prime.

5 Using indices

The index counts how many times the base multiplies: $5 \times 5 \times 5 = 5^3$.

$2^4 = 16 \quad 3^3 = 27 \quad 10^3 = 1000$

Watch out: $2^3 = 2 \times 2 \times 2 = 8$, not 6.

6 Prime decomposition

Use a factor tree until every branch ends in a prime.

$36 = 2^2 \times 3^2$

$90 = 2 \times 3^2 \times 5$

7 Squares and square roots

$7^2 = 49$, so $\sqrt{49} = 7$. They undo each other.

$\sqrt{144} = 12$

Perfect squares: 1, 4, 9, 16, 25, 36...

8 Number patterns

Find what happens each step: add a number, or multiply by one.

3, 7, 11, 15: add 4, next is **19**

2, 6, 18: multiply by 3, next is **54**

9 Spatial patterns

Count how much each new shape adds, then write a rule.

Squares from matchsticks: 4, 7, 10, so $\text{sticks} = 3n + 1$

10 squares: $3 \times 10 + 1 = 31$ sticks

10 Tables and rules

The step in the output tells you what to multiply by.

Input 1, 2, 3 and output 5, 8, 11: $y = 3x + 2$

$x = 10$: $y = 32$

11 The Cartesian plane

(x, y): across first, then up. The origin is $(0, 0)$.

$(3, 2)$ is 3 across and 2 up

Plot $y = x + 1$: $(0, 1), (1, 2), (2, 3)$